Real-Time Video Inference on 8× NVIDIA RTX PRO 5000

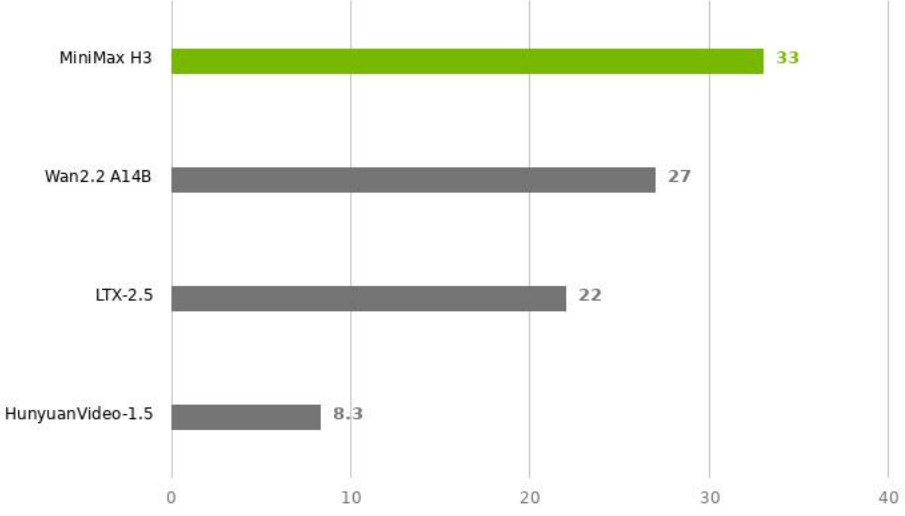
MiniMax H3 with vLLM-Omni

Shunyang Li | Public Technical Brief | October 2026



Real-Time 768p Raises the Systems Bar

For 362 frames at 24 FPS, real time requires end-to-end latency < 15.1 s (RTF < 1.0)



GENERATIVE BACKBONE · FULL PARAMETERS (B)

RTF = latency / media duration. The real-time deadline here is 15.1 s. H3 is the largest dense backbone in this peer set; Wan2.2 = 27B total / 14B active.



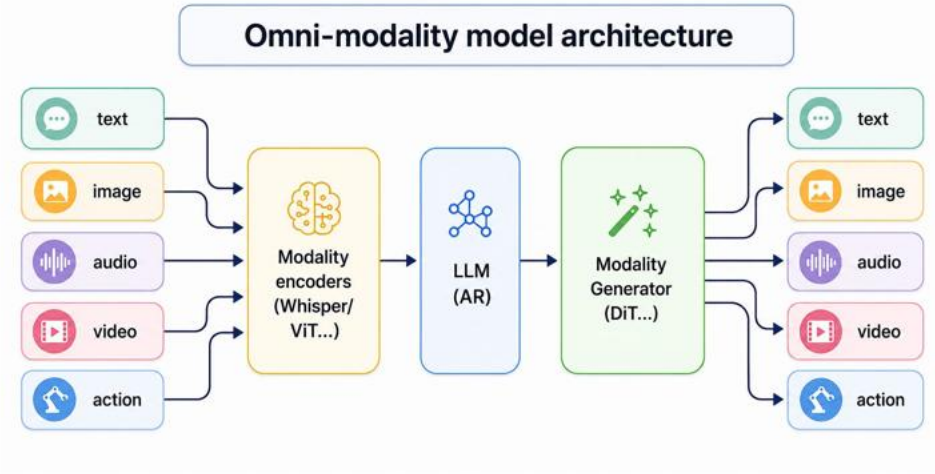
OPEN H3-BASE · NATIVE 768P · 33B DENSE

32B encoder → 33B dense Omni Transformer → Visual/Audio VAE. 50 blocks × 50 denoising steps; full attention. ≈130 GB raw BF16; ~13B AdaLN cacheable.

Sources: MiniMax H3, Wan2.2, LTX-2.5 and HunyuanVideo-1.5 official releases · author 362-frame workload

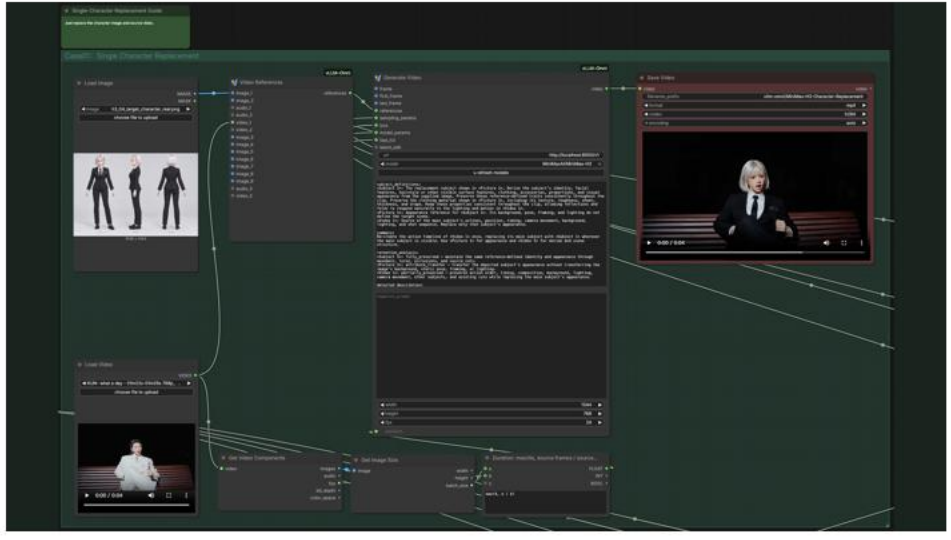
vLLM-Omni Adds Multi-GPU Serving to ComfyUI

ComfyUI authors workflows; vLLM-Omni serves them across multiple GPUs



Official vLLM-Omni Architecture

One serving layer spans multimodal inputs, autoregressive reasoning, and diffusion generation



ComfyUI Workflow · PR #9194

MiniMax H3 character replacement calls the vLLM-Omni Generate Video endpoint; PR #9194 is open

Sources: vLLM-Omni v0.26 public deck, slide 22 · vLLM-Omni PR #9194 (open)

Frontier Models · The Cards

SM120 compute is strong; PCIe-only scaling and sustained clocks set the eight-GPU envelope

SKU	CC / SMs	Memory	Power	Interconnect
RTX PRO 5000 Blackwell	12.0 110 SMs	72 GB GDDR7 ECC 48 GB variant	300 W public TBP 350 W local*	PCIe Gen5 ×16 No GPU NVLink
RTX PRO 6000 WS / Server	12.0 188 SMs	96 GB GDDR7 ECC	600 W	PCIe Gen5 ×16 No GPU NVLink
GeForce RTX 5090	12.0 170 SMs	32 GB GDDR7	575 W	PCIe Gen5 ×16 No NVLink
DGX Spark · GB10	12.1 48 SMs	128 GB unified LPDDR5X	140 W SoC 240 W system	NVLink-C2C in SoC 200 Gb/s ConnectX-7

Sources: NVIDIA specs + CUDA GPU table · local clock study: SwPowerCap 94.8–100%; −6.78%